GROUP NEXUS ENTERTAINMENT

DIGITAL PLATFORMS AND INFRASTRUCTURE - CT005H

DIGITAL TEAM HUB

COMPREHENSIVE DATABASE & COLLABORATION PLATFORM

INSTRUCTOR: ASSOC. PROF. NGUYEN THANH HAI, PHD

CAN THO, APRIL 6TH, 2026

OUR CORE TEAM

Full Name	Project Role
Tran Thi Minh Thao	Team Leader & Database Management
Nguyen Ngoc Tuong Vy	Data Analyst & Excel Expert
Tran Thi Ngoc Huyen	Network Infrastructure Engineer
Dang Bich Phuong	Digital Security & Ethics Specialist
Tieu Bui Nhu Y	Project Coordinator & Editor

PROJECT OVERVIEW

Digital Team Hub

- A system integrated with SQLite and Excel for team management over a local network
- Stores materials and tracks assignments

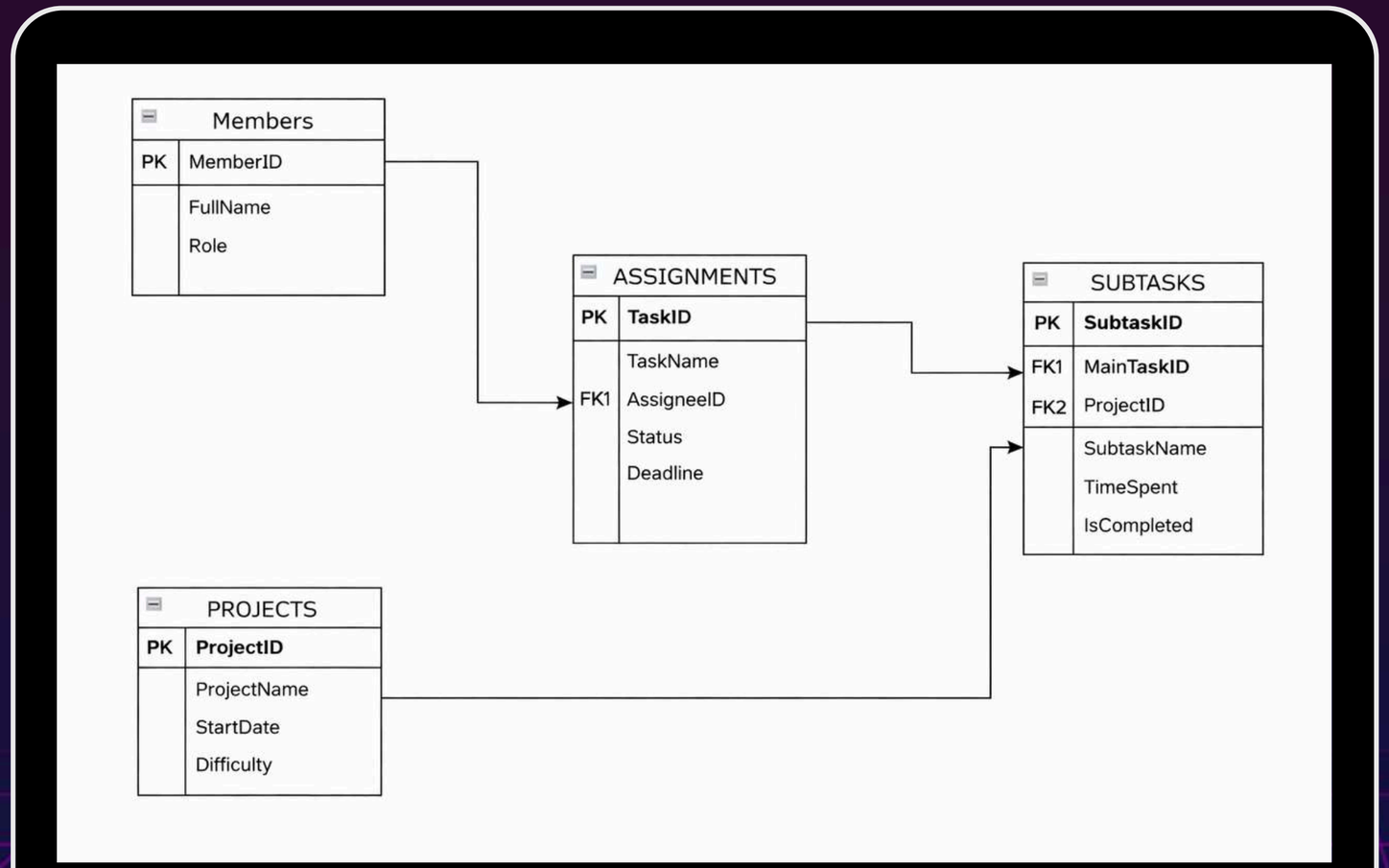
Goals

- Improve task management
- Support online collaboration
- Ensure data security

SYSTEM FEATURES

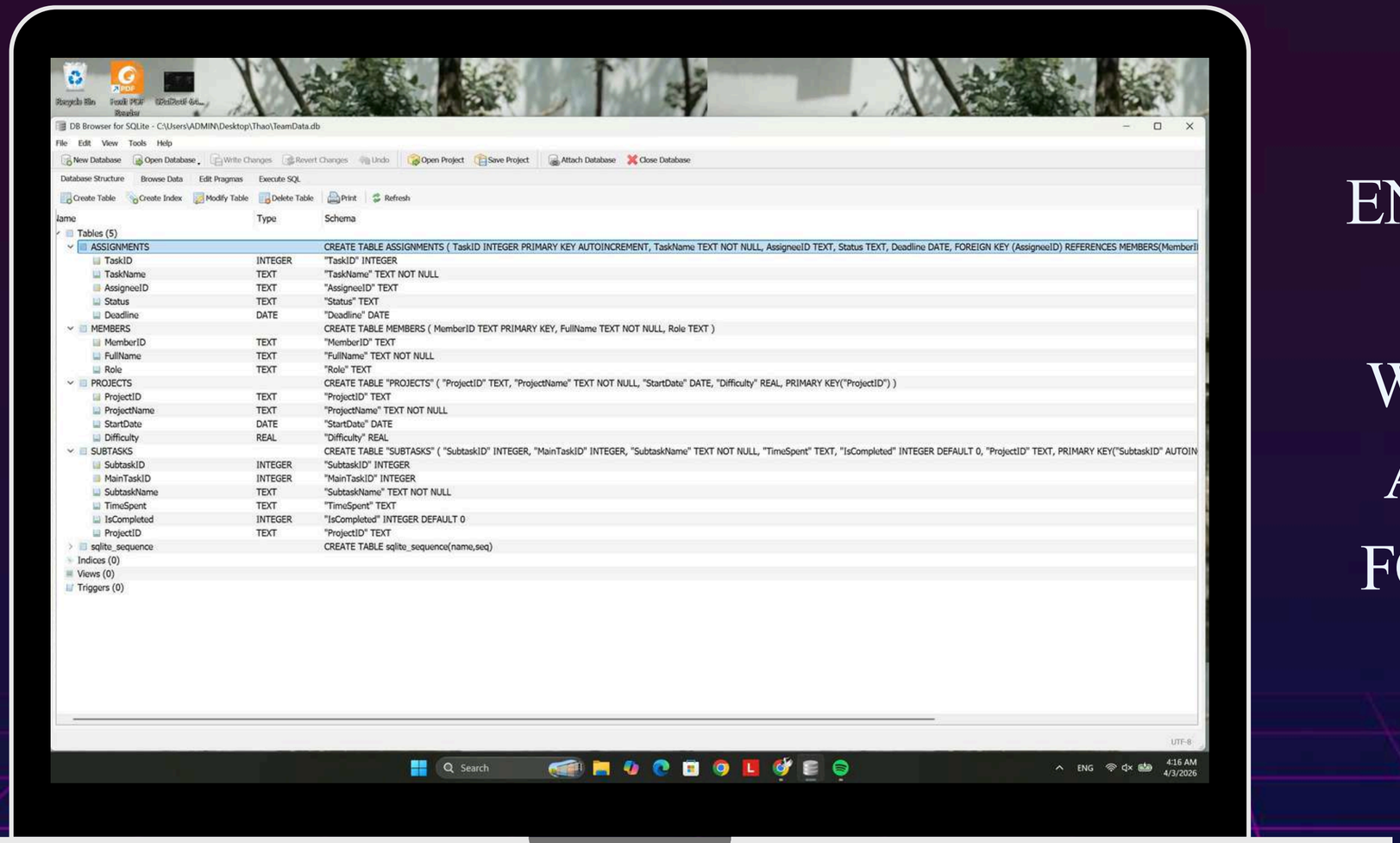
- PROJECT & TEAM CONTROL: CENTRALIZED MANAGEMENT OF PROJECTS AND PROFILES.
- SUBTASK DETAIL TRACKING: DEEP-LEVEL TASK BREAKDOWN AND MONITORING.
- SMART WORK ASSIGNMENT: EFFICIENT TASK DISTRIBUTION AND SCHEDULING.
- LIVE PROGRESS ANALYTICS: DATA-DRIVEN REPORTING WITH SQL AGGREGATES.

1. ENTITY-RELATIONSHIP MODEL OF THE SYSTEM



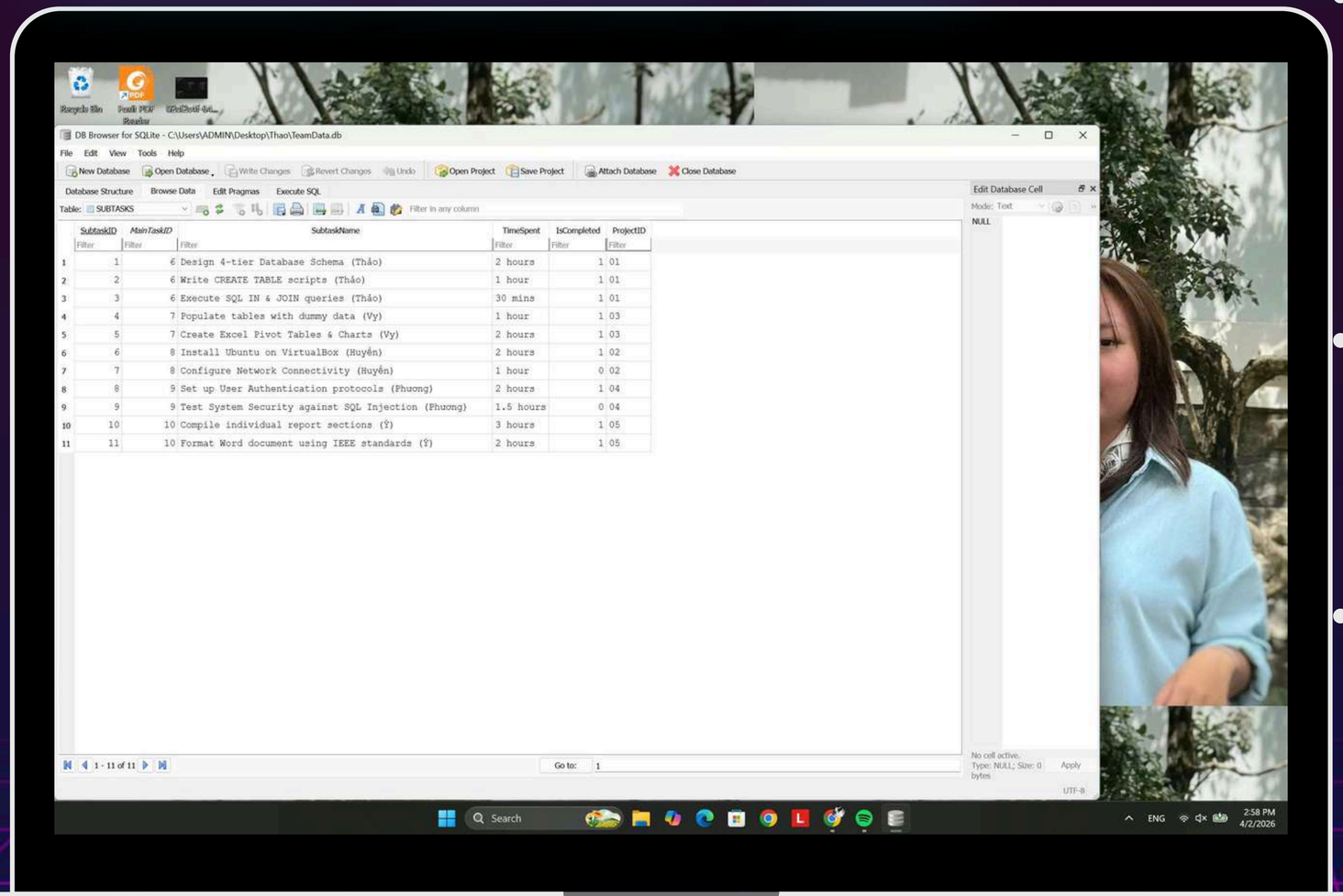
- PRIMARY KEYS (PK): UNIQUE IDENTIFIERS FOR EACH TABLE (MEMBERID, PROJECTID, TASKID, SUBTASKID).
- FOREIGN KEYS (FK): ESABLISHING RELATIONSHIPS BETWEEN TABLES TO ENSURE DATA INTEGRITY.
- RELATIONSHIPS: ONE MEMBER CAN BE ASSIGNED TO MULTIPLE TASKS.
- ONE PROJECT CAN CONTAIN MULTIPLE SUBTASKS.
- ATTRIBUTES: DETAILED DATA FIELDS SUCH AS PROJECTNAME, DEADLINE, AND ISCOMPLETED.

2. PHYSICAL DATABASE IMPLEMENTATION (SQLITE)



ENTITY RELATIONSHIP SCHEMA: DEFINED WITH PRIMARY KEYS AND FOREIGN KEYS FOR DATA INTEGRITY.

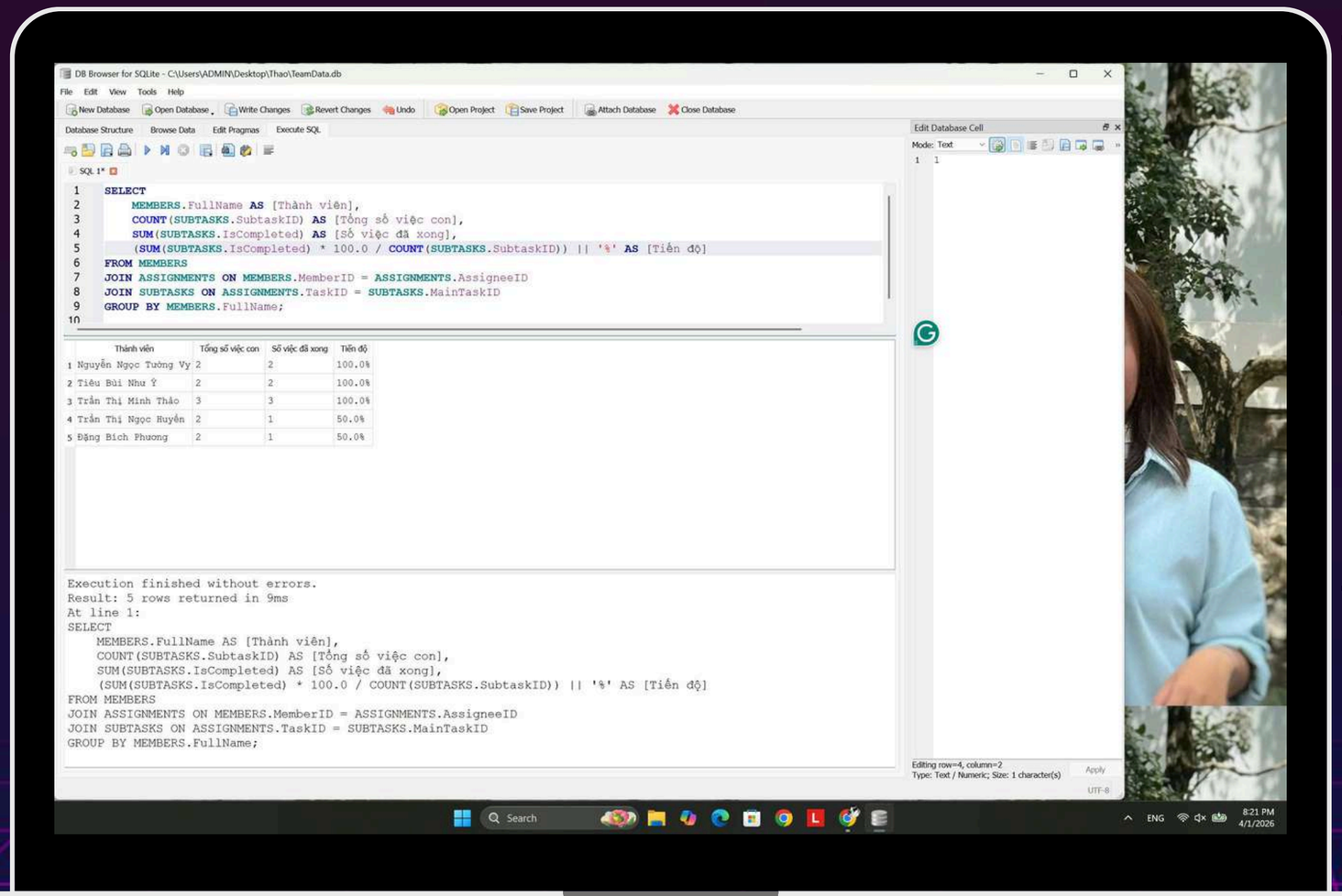
3. ACTUAL DATA SAMPLE: PROJECT & SUBTASK TRACKING (SUBTASKS TABLE)



	SubtaskID	MainTaskID	SubtaskName	TimeSpent	IsCompleted	ProjectID
1	1	6	Design 4-tier Database Schema (Thảo)	2 hours	1	01
2	2	6	Write CREATE TABLE scripts (Thảo)	1 hour	1	01
3	3	6	Execute SQL IN & JOIN queries (Thảo)	30 mins	1	01
4	4	7	Populate tables with dummy data (Vy)	1 hour	1	03
5	5	7	Create Excel Pivot Tables & Charts (Vy)	2 hours	1	03
6	6	8	Install Ubuntu on VirtualBox (Huyền)	2 hours	1	02
7	7	8	Configure Network Connectivity (Huyền)	1 hour	0	02
8	8	9	Set up User Authentication protocols (Phuong)	2 hours	1	04
9	9	9	Test System Security against SQL Injection (Phuong)	1.5 hours	0	04
10	10	10	Compile individual report sections (Ỹ)	3 hours	1	05
11	11	10	Format Word document using IEEE standards (Ỹ)	2 hours	1	05

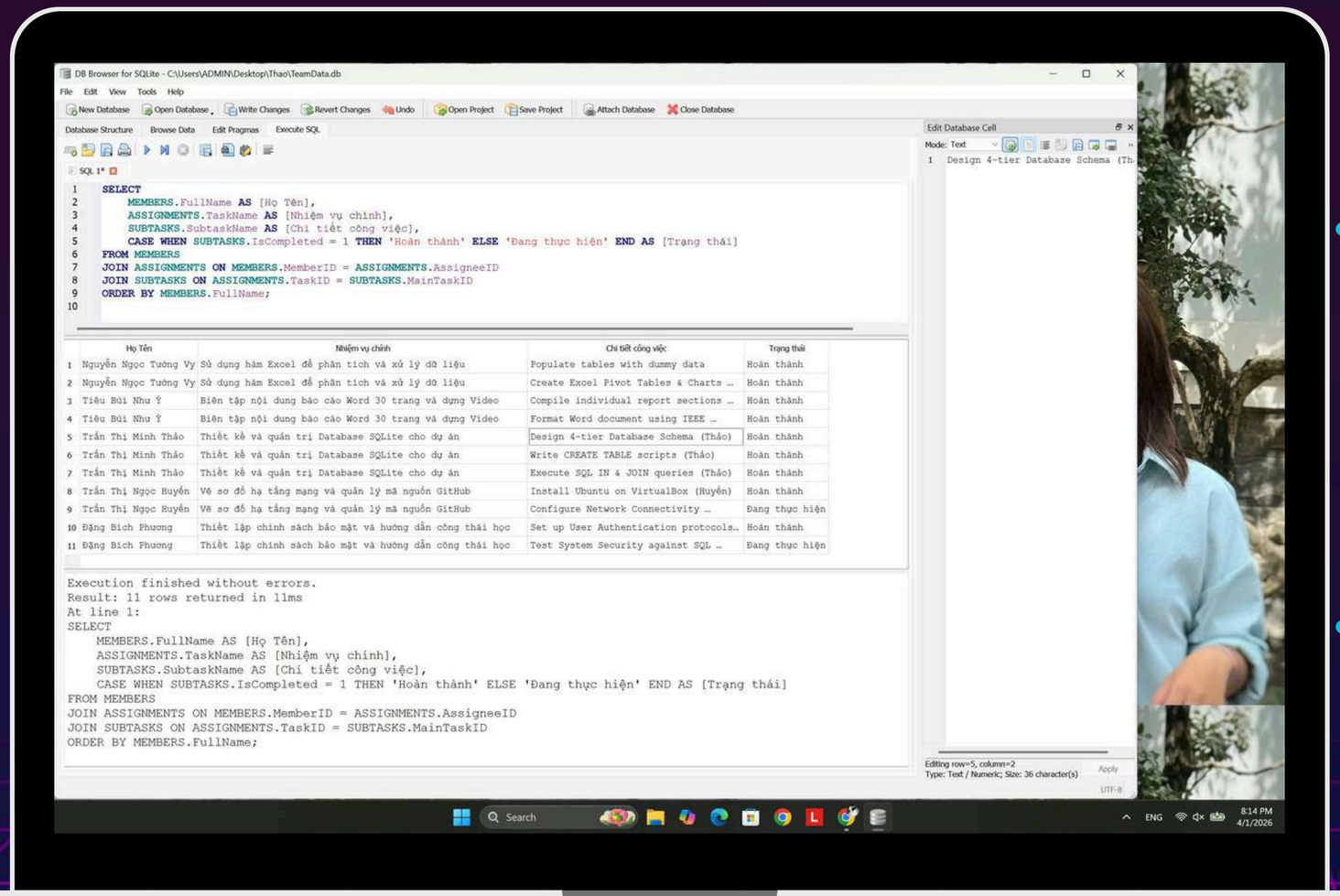
- **POPULATED DATA:** THE SUBTASKS TABLE CONTAINS SAMPLE RECORDS DEMONSTRATING THE CONNECTION BETWEEN PROJECTS (PROJECTID) AND SPECIFIC TASKS.
- **KEY FIELDS FOR REPORTING:** THIS VIEW HIGHLIGHTS THE KEY DATA POINTS REQUIRED FOR TASK-LEVEL PROGRESS REPORTS.
- **TASK STATUS:** INCLUDES ISCOMPLETED STATUS (1 FOR COMPLETE, 0 FOR ONGOING) FOR DETAILED PROGRESS TRACKING.

4. TEAM PROGRESS ANALYTICS (SQL AGGREGATE FUNCTIONS)



- **PERFORMANCE ANALYTICS:** TRACKING TOTAL SUBTASKS AND COMPLETION RATES USING COUNT AND SUM FUNCTIONS.
- **REAL-TIME PROGRESS:** CALCULATING INSTANT WORK PERCENTAGES WITH THE (SUM / COUNT) * 100 FORMULA.
- **WORKLOAD INSIGHT:** IDENTIFYING MEMBER PROGRESS AND TASK DISTRIBUTION FOR BETTER TEAM MANAGEMENT.

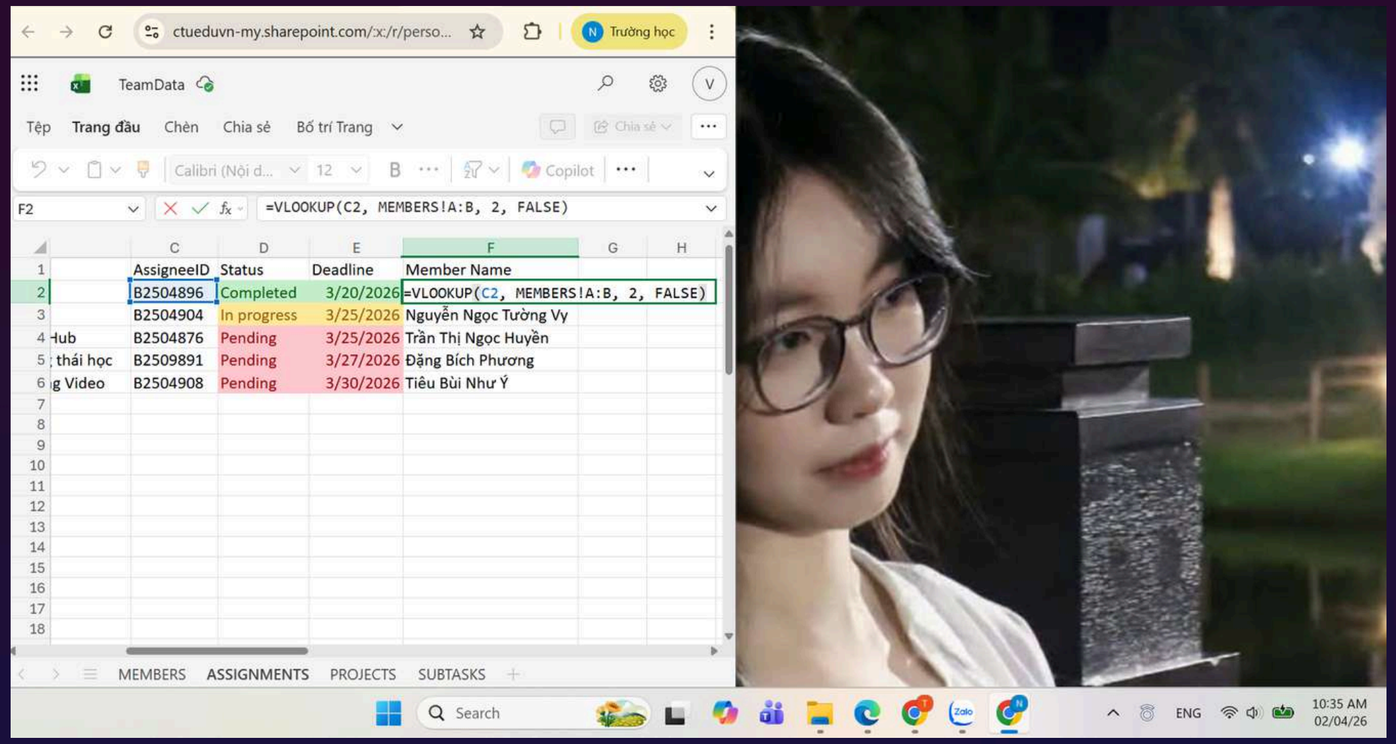
5. ADVANCED DATA JOINS & TASK MAPPING



	Họ Tên	Nhiệm vụ chính	Chi tiết công việc	Trạng thái
1	Nguyễn Ngọc Tường Vy	Sử dụng hàm Excel để phân tích và xử lý dữ liệu	Populate tables with dummy data	Hoàn thành
2	Nguyễn Ngọc Tường Vy	Sử dụng hàm Excel để phân tích và xử lý dữ liệu	Create Excel Pivot Tables & Charts …	Hoàn thành
3	Tiêu Bùi Như Ý	Biên tập nội dung báo cáo Word 30 trang và dựng Video	Compile individual report sections …	Hoàn thành
4	Tiêu Bùi Như Ý	Biên tập nội dung báo cáo Word 30 trang và dựng Video	Format Word document using IEEE …	Hoàn thành
5	Trần Thị Minh Thảo	Thiết kế và quản trị Database SQLite cho dự án	Design 4-tier Database Schema (Thảo)	Hoàn thành
6	Trần Thị Minh Thảo	Thiết kế và quản trị Database SQLite cho dự án	Write CREATE TABLE scripts (Thảo)	Hoàn thành
7	Trần Thị Minh Thảo	Thiết kế và quản trị Database SQLite cho dự án	Execute SQL IN & JOIN queries (Thảo)	Hoàn thành
8	Trần Thị Ngọc Huyền	Vẽ sơ đồ hạ tầng mạng và quản lý mã nguồn GitHub	Install Ubuntu on VirtualBox (Huyền)	Hoàn thành
9	Trần Thị Ngọc Huyền	Vẽ sơ đồ hạ tầng mạng và quản lý mã nguồn GitHub	Configure Network Connectivity …	Đang thực hiện
10	Đặng Bích Phương	Thiết lập chính sách bảo mật và hướng dẫn công thái học	Set up User Authentication protocols…	Hoàn thành
11	Đặng Bích Phương	Thiết lập chính sách bảo mật và hướng dẫn công thái học	Test System Security against SQL …	Đang thực hiện

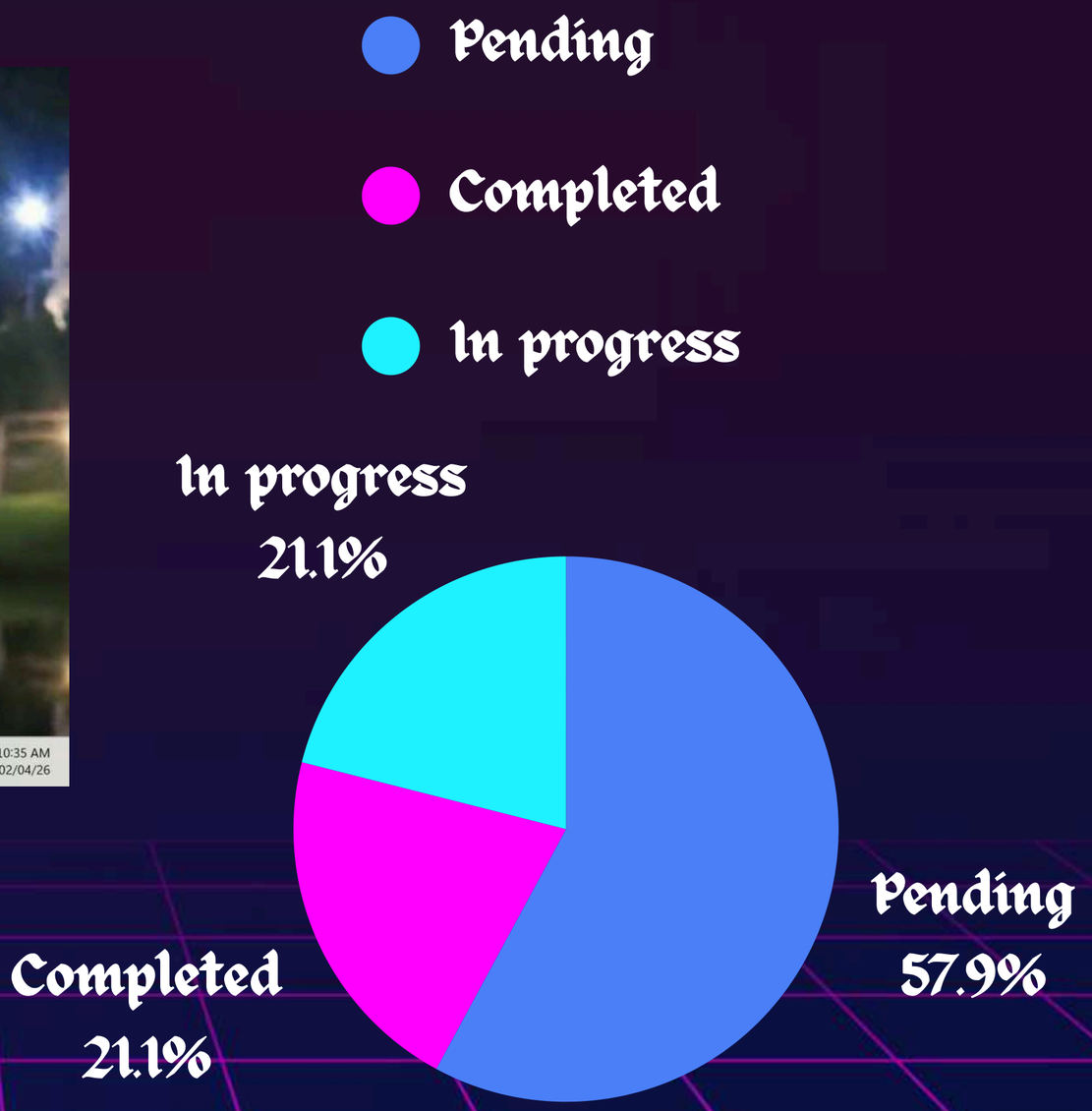
- **MULTI-TABLE INTEGRATION:** SUCCESSFULLY JOINING MEMBERS, ASSIGNMENTS, AND SUBTASKS TO CREATE A UNIFIED VIEW.
- **CONDITIONAL LOGIC:** IMPLEMENTING CASE WHEN STATEMENTS TO TRANSFORM RAW BITS (0/1) INTO READABLE STATUS: "HOÀN THÀNH" OR "ĐANG THỰC HIỆN".
- **DETAILED VISIBILITY:** [illegible]ARLY MAPPING EACH SUBTASK BACK TO ITS PRIMARY ASSIGNMENT AND THE RESPONSIBLE MEMBER.

6. DATA ANALYSIS & SPREADSHEET INTEGRATION

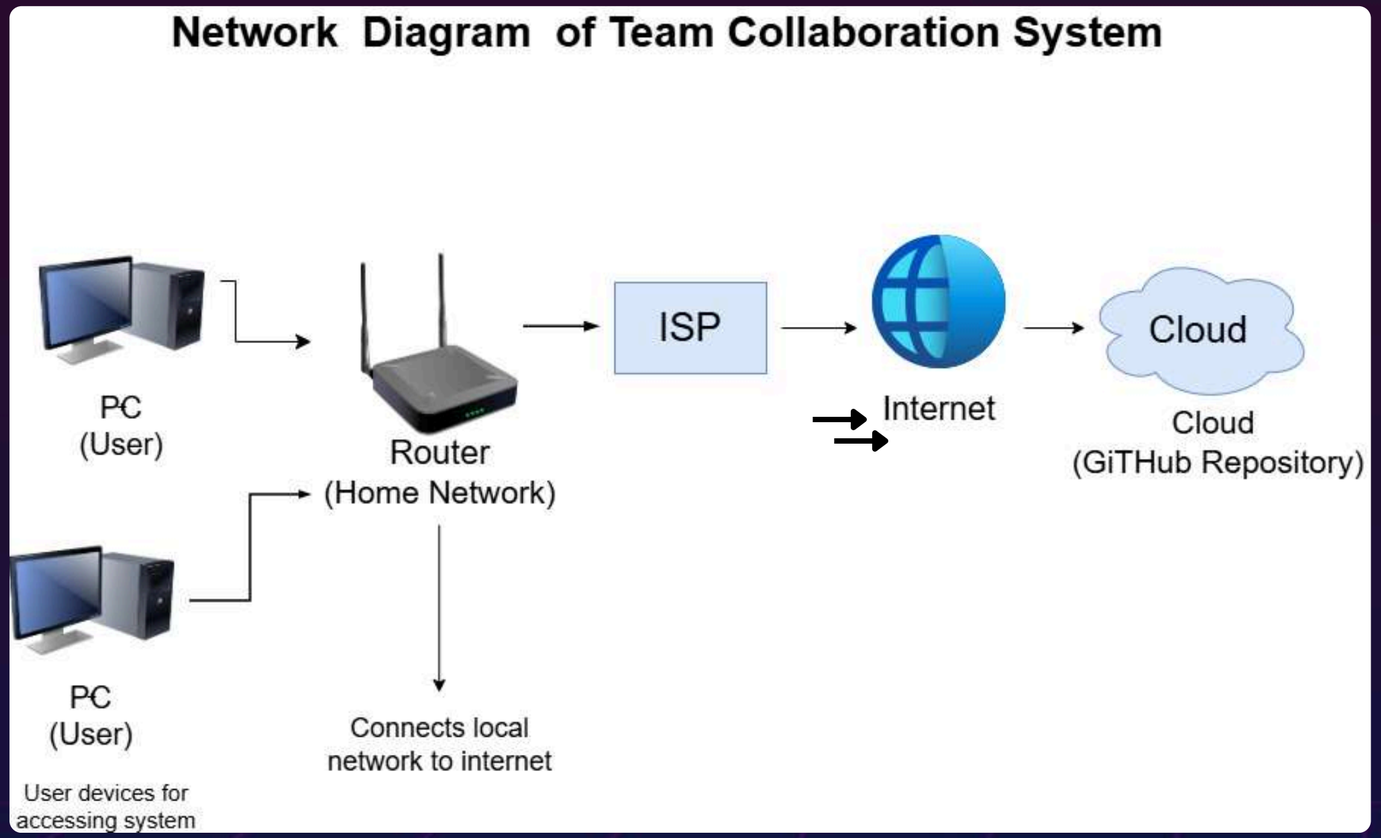


	AssigneeID	Status	Deadline	Member Name
	B2504896	Completed	3/20/2026	=VLOOKUP(C2, MEMBERS!A:B, 2, FALSE)
	B2504904	In progress	3/25/2026	Nguyễn Ngọc Tường Vy
Hub	B2504876	Pending	3/25/2026	Trần Thị Ngọc Huyền
thái học	B2509891	Pending	3/27/2026	Đặng Bích Phương
g Video	B2504908	Pending	3/30/2026	Tiêu Bùi Như Ý

DATA MANAGEMENT & AUTOMATION WITH VLOOKUP

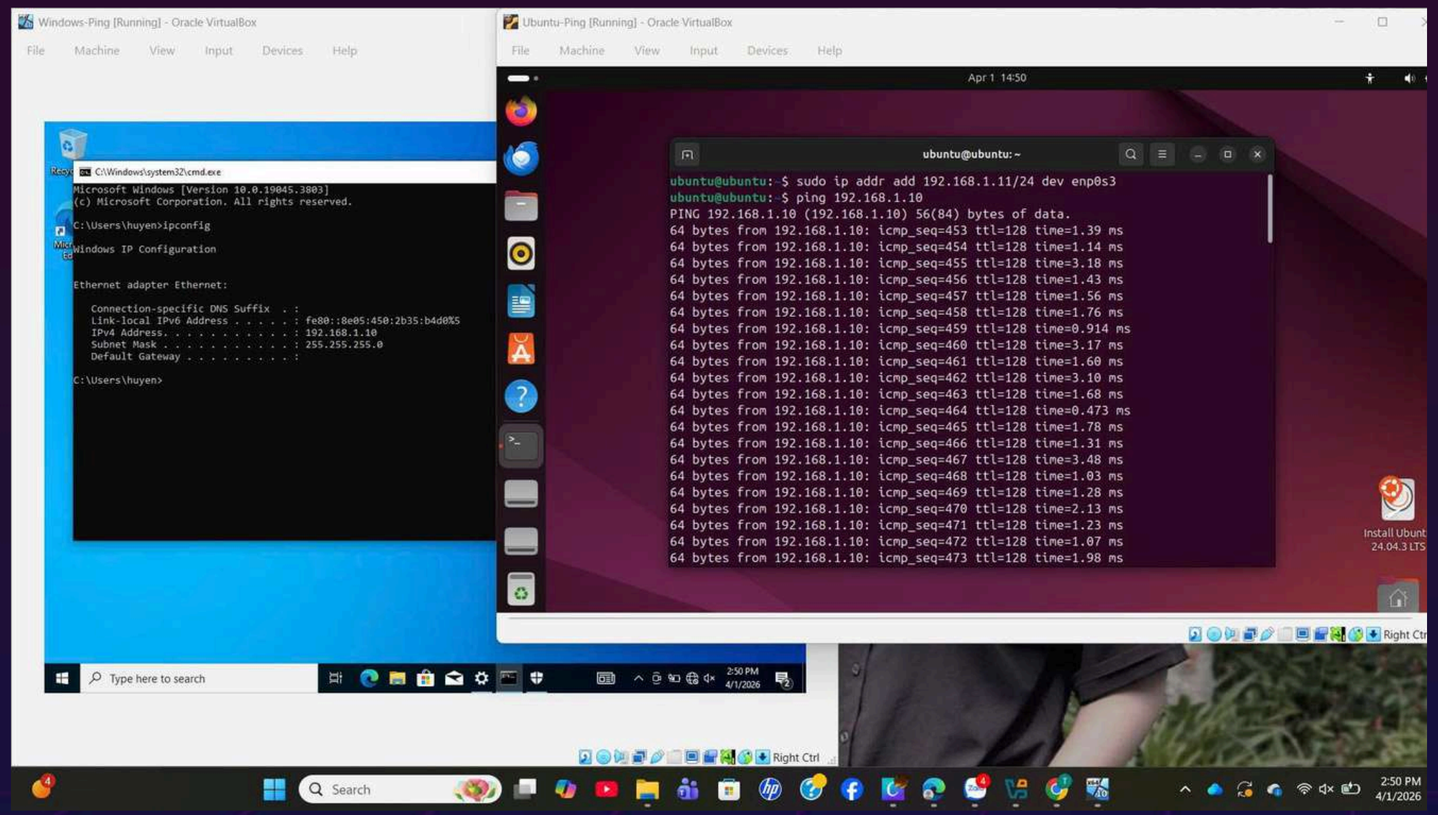


7. SYSTEM INFRASTRUCTURE & SECURITY PROTOCOLS



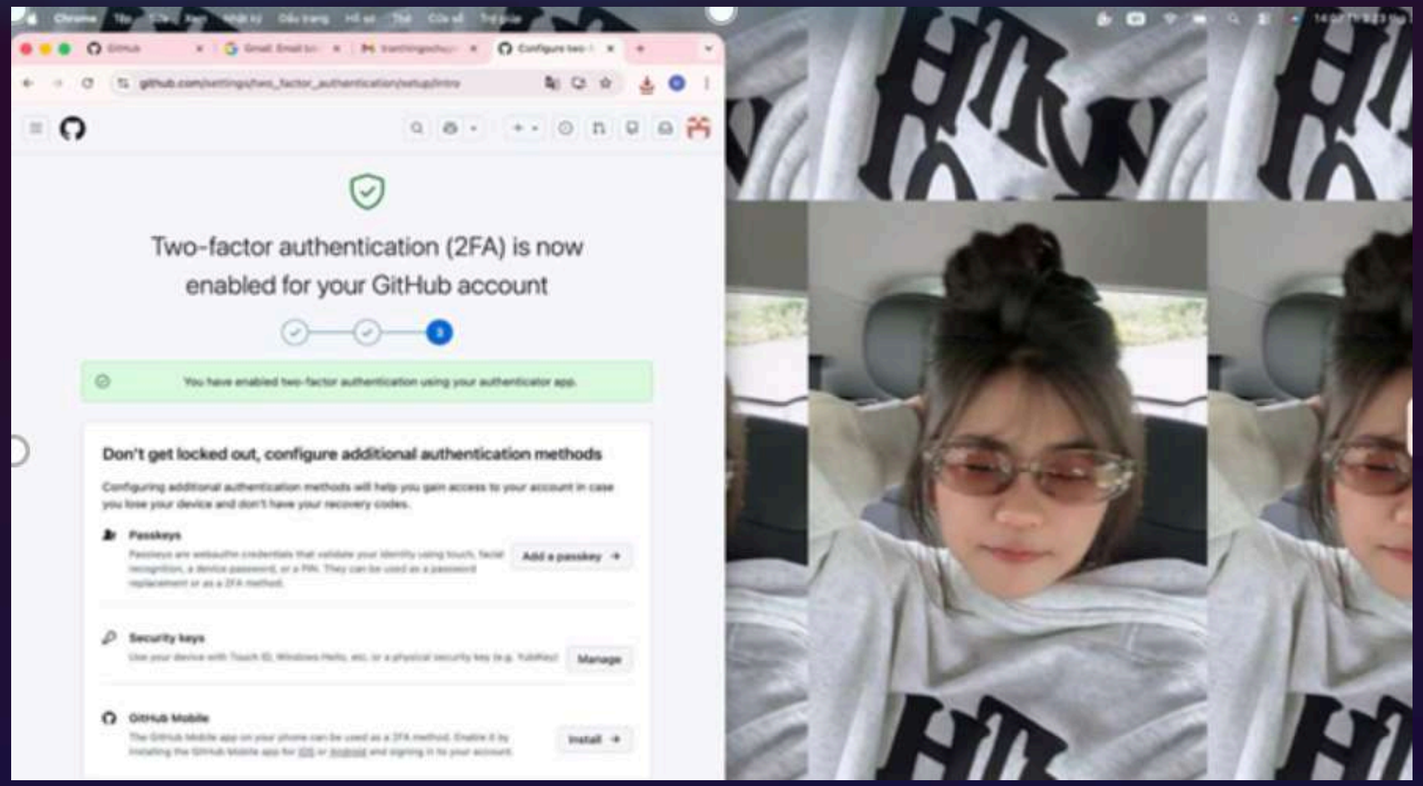
- **HOME NETWORK SETUP:** LAN CONNECTION VIA ROUTER.
- **ISP & INTERNET:** GATEWAY TO GLOBAL CONNECTIVITY.
- **CLOUD COLLABORATION:** VERSION CONTROL AND STORAGE VIA GITHUB REPOSITORY.
- **PROTOCOL:** TCP/IP FOR STABLE DATA TRANSMISSION.

7. SYSTEM INFRASTRUCTURE & SECURITY PROTOCOLS



- **CONNECTIVITY PROTOCOL:** TC
- **COLLABORATION TOOL:** GITHU DESKTOP & GIT COMMANDS.
- **STATUS:** SUCCESS (VERIFIED VIA CLOUD SYNC).
- **DATA FLOW:** LOCAL PC → HOME ROUTER → INTERNET → GITHUB CLOUD.

8. SECURITY POLICIES & ATTACK PREVENTION

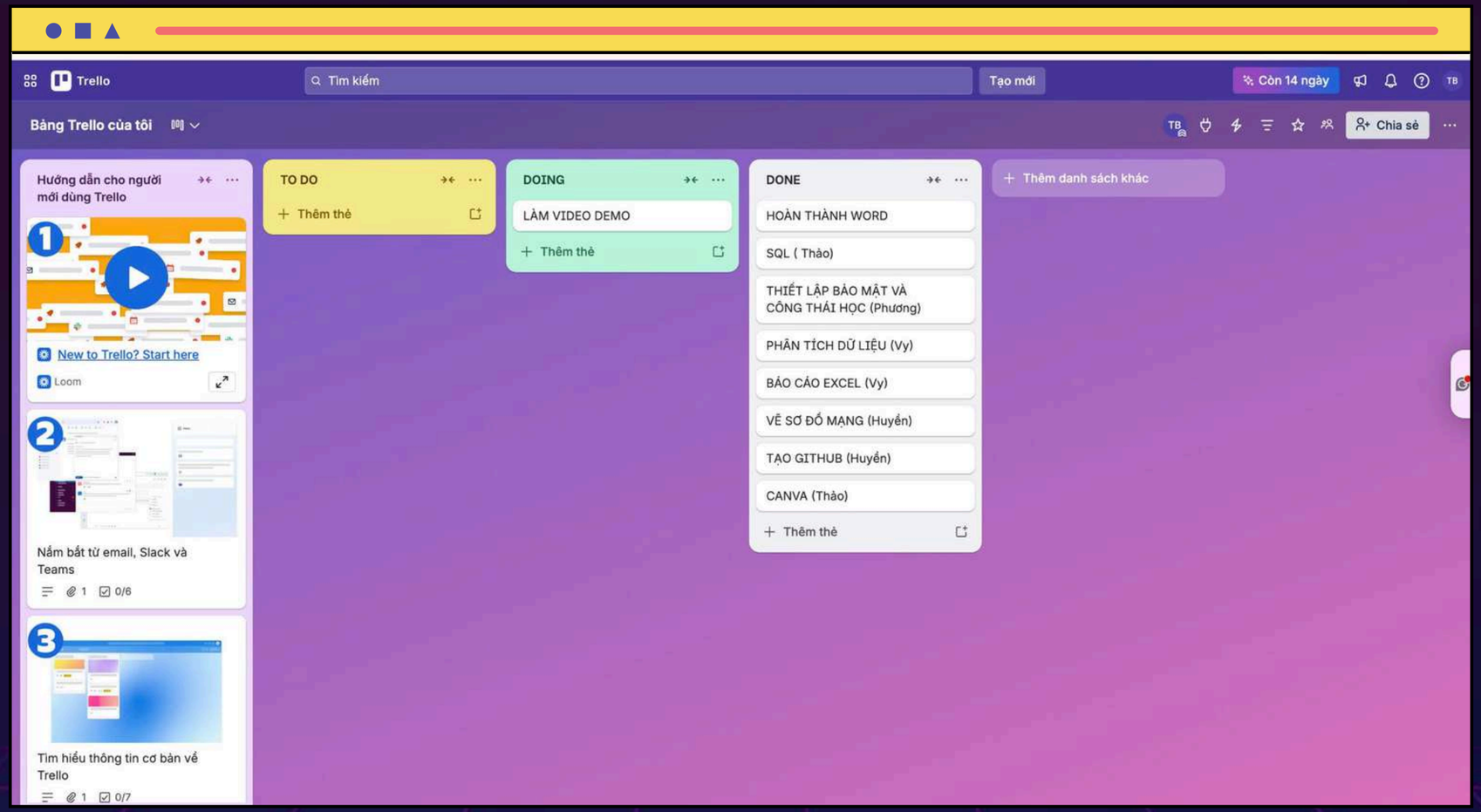


- ACCOUNT SECURITY: IMPLEMENTATION OF TWO-FACTOR AUTHENTICATION (2FA) FOR GITHUB REPOSITORY.
- ACCESS CONTROL: ONLY AUTHORIZED TEAM MEMBERS CAN ACCESS AND MODIFY PROJECT DATA.
- SOFTWARE ETHICS: 100% USAGE OF LICENSED AND OPEN-SOURCE TOOLS (VIRTUALBOX, SQLITE, CANVA).

9. AI PROMPT ENGINEERING & WORKFLOW

- PROMPT 1 (DATABASE ARCHITECTING):
- "SUGGEST A DATABASE SCHEMA FOR A TEAM MANAGEMENT SYSTEM INCLUDING TABLES FOR MEMBERS AND ASSIGNMENTS WITH PRIMARY AND FOREIGN KEYS."
- PROMPT 2 (COMPLEX QUERY SYNTHESIS):
- "GENERATE AN SQL QUERY TO JOIN THE 'MEMBERS' AND 'ASSIGNMENTS' TABLES TO DISPLAY FULL NAMES AND THEIR TASK STATUS."
- PROMPT 3 (DATA OPTIMIZATION):
- "GUIDE ON USING VLOOKUP AND CONDITIONAL FORMATTING TO TRACK PROJECT DEADLINES IN EXCEL."
- PROMPT 4 (HEALTH & SECURITY POLICY):
- "DEFINE DIGITAL HEALTH AND ERGONOMICS PRINCIPLES FOR IT STUDENTS WORKING LONG HOURS ON COMPUTERS."

10. COLLABORATIVE WORKFLOW & TEAM COORDINATION



GROUP PROJECT WORKFLOW AND TASK MANAGEMENT ON TRELLO

THANKS FOR LISTENING

Cảm ơn tất cả

thành viên

của nhóm

NEXUS

đã cố gắng